

Workshop No. 4

Student Mental Health Support Platform



**UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS**

Team Members

Alan Santiago Agudelo Sarmiento – 20222020170

Juan David Cardozo Trujillo - 20231020155

Julian Ernesto Romero Gutierrez - 20231020164

Nicolas Alexander Sierra Contreras - 20222020197

Professor

Eng. Carlos Andrés Sierra, M.Sc.

Universidad distrital Francisco José de Caldas

Engineering faculty

Systems Analysis & Design

2026-I

Executive Summary

General summary

Workshop 4 transforms the architectural design and previous findings into empirical evidence through computational simulations of the **Student Mental Health Support Platform**. The central objective is to validate design decisions, quantify operational performance, and explore emergent behaviors under varied conditions. Simulations focus on the system's critical processes: registration, authentication, profile generation, matching with peer counselors, and metrics related to availability, load, and trust.

Objectives

- Validate the core modules of the system and their interactions.
- Evaluate performance under four standardized scenarios: **Baseline**, **Midterm/Finals Stress**, **Counselor Shortage**, and **Trust Collapse**.
- Quantify operational metrics: match rate, unmatched rate, counselor load distribution, number of counselors in burnout, queue lengths, and estimated response times.
- Investigate complex dynamics: feedback loops, nonlinear responses, and sensitivity to critical parameters (trust, demand peaks, resource availability).
- Produce actionable recommendations for pilot deployment, resource planning, and risk mitigation.

Methodology

- **Approach:** coordinated set of independent simulations implemented in **Python**, with analysis and visualization using *numpy*, *pandas*, and *matplotlib*.
- **Shared parameters:**

NUM_STUDENTS = 500
NUM_PEER_COUNSELORS = 30
PEAK_MULTIPLIER = 2.5
AVG_RESPONSE_TIME = 5 min
BURNOUT_THRESHOLD = 20 sessions
TRUST_LEVEL = 0.85.

- **Scenarios:** the four scenarios are applied consistently to each model to ensure comparability.
- **Validation:** parameter calibration with prior survey data, multi-seed execution to report mean $\pm$ standard deviation, and sensitivity analysis for key parameters.
- **Deliverables:** reproducible scripts, CSV tables with per-scenario metrics, standardized plots (titles, labeled axes, legends), and technical documentation in English.

Model design and assumptions

- **Paradigm:** discrete-event simulation to model arrivals, authentication, and resource assignment.
- **Modeled components:** student arrival events, probabilistic login (trust model), immediate profile creation, greedy lowest-load matching policy, session counting and burnout enforcement.
- **Key assumptions:** profile creation is instantaneous after successful login; session duration is simplified or not modeled in detail; no frontend; students do not automatically retry after a failed login in the base model.
- **Scope limits:** detailed real-time messaging, advanced appointment scheduling, and full professional counselor workflows are out of scope for this simulation set.

Expected results and main findings

- **Baseline:** high match rate, average counselor loads below burnout threshold, and response times near the 5-minute target (estimated).
- **Midterm/Finals Stress:** demand surges (modeled via **PEAK_MULTIPLIER**) cause significant increases in queues and counselor loads; congestion emerges and response times and unmatched rates grow nonlinearly.
- **Counselor Shortage:** reduced counselor availability leads to disproportionate increases in waiting and unmatched students, showing high sensitivity to human resource capacity.

- **Trust Collapse:** lower effective **TRUST_LEVEL** reduces logins and matches, causing underutilization of available counselors and demonstrating that perceived privacy and anonymity are critical for adoption.

Sensitivity analysis and emergent behavior

- The system exhibits nonlinear responses to changes in **PEAK_MULTIPLIER** and **TRUST_LEVEL**; small variations in trust or capacity can trigger abrupt increases in queues and burnout.
- There are tipping points where the combination of high demand and low availability produces cascade effects (more abandonments → lower trust → lower usage).
- Aggregate metrics (match rate, percentage of burned-out counselors) identify operational thresholds for activating mitigation protocols.

Operational and design implications

- Implement surge response protocols: on-call rotations, counselor reserves, and automatic escalation mechanisms.
- Prioritize low-friction, privacy-preserving authentication and clear consent flows to maintain trust and reduce login failures.
- Enforce workload caps and mandatory rest periods to prevent burnout; deploy real-time monitoring of key indicators.
- Define SLAs based on simulated metrics (target response times, maximum queue lengths) and design pilots that compare real behavior with simulation outputs.

Recommendations for pilot phase and next steps

- Run controlled pilots to collect real usage data for model calibration and validation.
- Deploy monitoring dashboards to detect early signs of overload and trust erosion.
- Prioritize UX and privacy guarantees before scaling the service.
- Conduct complementary analyses focused on appointment scheduling, chat dynamics, and escalation paths to professional care.

Conclusion:

Workshop 4 delivers a rigorous, quantitative evaluation of the **Student Mental Health Support Platform** through reproducible simulations and standardized scenarios. The results validate the proposed architecture, reveal critical operational vulnerabilities (burnout, counselor shortages, trust erosion), and prioritize technical and organizational interventions. The generated evidence forms a solid basis for pilot design, operational policy, and ethically informed scaling decisions.

System Model Development

Objective

Design and implement a reproducible **discrete-event simulation** of the Student Mental Health Support Platform that models the initial user flow — **registration** → **authentication** → **profile creation** → **matching** — and produces actionable metrics to evaluate operational performance and risks across four standardized scenarios. The model must be parameterized, seedable for reproducibility, and produce CSV and PNG outputs for analysis.

System Architecture Translation

Goal

Map the three architectural layers (Presentation, Application, Infrastructure) into concrete simulation components so each architectural responsibility is traceable to code, inputs, and outputs.

Mapping

Architecture Layer	Simulation component	Responsibilities and outputs
Presentation	Event generator (student_arrival, registration_attempt, login_attempt)	Generates demand; models front-end friction via probabilistic login; output: arrival stream and login success/failure counts
Application	Authentication module; Profile service; Matching System; Metrics collector	Implements attempt_login(trust_level); immediate profile creation; greedy lowest-load matching; aggregates per-scenario metrics and sanity checks
Infrastructure	Peer Counselor pool; persistence layer (CSV); monitoring counters	Models counselor availability and burnout; persists results and plots; output: counselor load arrays, burned-out counts, exported CSVs and PNGs

Design principles

- **Modularity:** each architectural responsibility is a small function or data structure (e.g., attempt_login(), assign_counselor(), counselor_pool).
- **Parameterization:** scenario modifiers are passed to the application layer so the same core simulation runs all scenarios.
- **Observability:** metrics and logs are collected at each layer to enable calibration and debugging.

System Entities and Relationships

Entities and attributes

Entity	Key attributes	Purpose
Student	id; arrival_index; login_success (bool); profile_created (bool); matched (bool)	Demand generator; one-shot interaction in base model
Peer Counselor	id; sessions_completed (int); available (bool)	Service resource; becomes unavailable when burned out

Matching System	match_policy; metrics; queue (logical)	Assigns counselors and records outcomes
Environment	NUM_STUDENTS; NUM_PEER_COUNSELORS; TRUST_LEVEL; PEAK_MULTIPLIER; BURNOUT_THRESHOLD; AVG_RESPONSE_TIME	Global parameters and scenario modifiers

Primary relationships and interactions

- After a successful login and profile creation, a **Student** issues a request_matching to the **Matching System**.
- The **Matching System** selects a **Peer Counselor** according to the match_policy (lowest sessions completed) and increments that counselor's sessions_completed.
- When a **Peer Counselor** reaches BURNOUT_THRESHOLD, they transition to available = False and are excluded from future assignments.
- The global **TRUST_LEVEL** affects login_success probability; scenario modifiers change arrival intensity or counselor availability.

State transition summary

- **Student:** arrived → login_success or login_failed → if success → profile_created → matched or unmatched.
- **Peer Counselor:** available → in_session → available (increment sessions) → unavailable when burned out.

Simulation Paradigm Selection

Chosen paradigm

Discrete-Event Simulation (DES) implemented as a seeded, deterministic loop over a finite student population, with optional extension to an explicit event queue if time-series metrics are required.

Why DES

- The system is driven by discrete events (arrivals, login attempts, matching assignments, session completions). DES models these naturally and efficiently.
- DES captures resource contention, queueing effects, and threshold phenomena (burnout) without unnecessary continuous dynamics.
- DES is straightforward to implement in Python and easy to instrument for the required metrics.

Alternatives considered

- **Agent-Based Modeling:** would allow richer individual behaviors (retries, scheduling) but increases complexity and runtime without immediate benefit for the initial flow.
- **System Dynamics:** useful for high-level policy exploration but lacks the event granularity needed to evaluate matching policies and counselor assignment logic.

Implementation notes

- Implement as a loop over num_students with seeded randomness (random.seed(seed)), or use an event queue if modeling timestamps and service durations.
- Keep functions small and testable: attempt_login(trust_level), select_lowest_load(counselors), run_scenario(params, seed).

Simulation Boundaries and Scope

In scope

- **Flows:** registration, probabilistic authentication (TRUST_LEVEL), immediate profile creation, matching to peer counselors.
- **Resources:** Peer Counselor pool with session counting and burnout enforcement.
- **Scenarios:** Baseline; Midterm/Finals Stress (increase arrivals via PEAK_MULTIPLIER); Counselor Shortage (reduced counselor pool); Trust Collapse (reduced TRUST_LEVEL).
- **Outputs:** per-seed CSV with metrics, summary CSV with mean $\pm$ std, PNG plots (Matched vs Unmatched; Counselor Workload Distribution), and a sensitivity CSV.
- **Validation:** multi-seed runs (recommended 5 seeds), sanity checks (Matched + Unmatched == NumStudents), and sensitivity analysis for TRUST_LEVEL and PEAK_MULTIPLIER.

Out of scope

- Frontend/UI, real chat session dynamics, advanced appointment scheduling, professional counselor escalation workflows, detailed session duration modeling beyond simple proxies, and external system integrations.

Key assumptions

- Profile creation is instantaneous after successful login.
- Students do not retry after a failed login in the base model.
- Session duration is simplified; AVG_RESPONSE_TIME is used as an operational proxy unless explicit service times are modeled.
- Burnout is modeled as a simple session count threshold (BURNOUT_THRESHOLD).

Required metrics and validation checks

- **Primary metrics:** Matched, Unmatched, MatchedRate, AvgCounselorLoad, NumBurnedOut, EstResponseTime (proxy).
- **Per-seed CSV columns:** Scenario, Seed, NumStudents, Matched, Unmatched, MatchedRate, AvgCounselorLoad, NumBurnedOut, EstResponseTime.
- **Summary CSV:** mean $\pm$ std for MatchedRate, NumBurnedOut, AvgLoad, EstResponseTime.
- **Sanity checks:** assert Matched + Unmatched == NumStudents for every run.
- **Sensitivity checks:** run Baseline with TRUST_LEVEL $\pm$ 0.05 and PEAK_MULTIPLIER $\pm$ 0.5 and report deltas in MatchedRate and NumBurnedOut.

Implementation Checklist and Next Steps

1. **Parameterize** the simulation with the exact shared values: NUM_STUDENTS = 500, NUM_PEER_COUNSELORS = 30, PEAK_MULTIPLIER = 2.5, AVG_RESPONSE_TIME = 5, BURNOUT_THRESHOLD = 20, TRUST_LEVEL = 0.85.
2. **Implement core functions:** attempt_login(), select_lowest_load(), run_single_simulation(), run_experiments() with multi-seed support.
3. **Export artifacts:** per-seed CSV, summary CSV, sensitivity CSV, and PNG plots for each scenario.
4. **Run validation:** multi-seed runs (≥ 5), sanity checks, and sensitivity analysis.
5. **Document** assumptions, limitations, and how simulated components map to the architecture.
6. **Interpret results:** produce Results and Analysis sections that translate metrics into operational recommendations (surge protocols, privacy priorities, workload caps).

User Flow Simulation

Goal

Model the end-to-end student journey: **arrival** → **registration** → **authentication** → **profile creation** → **matching** with peer counselors. The simulation must be reproducible, scenario-driven, and produce per-run CSVs, aggregated summaries, and visualizations for analysis.

Global parameters (use exactly)

- **NUM_STUDENTS = 500**
- **NUM_PEER_COUNSELORS = 30**
- **PEAK_MULTIPLIER = 2.5**
- **AVG_RESPONSE_TIME = 5** minutes (proxy)
- **BURNOUT_THRESHOLD = 20** sessions
- **TRUST_LEVEL = 0.85**

Scenarios to implement

1. Baseline Scenario
2. Midterm Finals Stress Scenario
3. Counselor Shortage Scenario
4. Trust Collapse Scenario

Execution control

- Seeds for reproducibility: **[42, 43, 44, 45, 46]**
- Outputs per run: CSV row with metrics; aggregated summary CSV; PNG plots (if plotting library available).

Registration Simulation:

Purpose

Simulate student arrivals and registration attempts; produce the cohort that proceeds to authentication.

Function name

run_registration

Inputs

- num_students
- registration_failure_rate (default 0.0)
- RNG seeded externally

Behavior

- For each arriving student perform a Bernoulli trial with probability $1 - \text{registration_failure_rate}$.
- Successful trials append student ID to registered. Failed trials increment num_failed.

Outputs

- registered (list of IDs)
- num_registered (int)
- num_failed (int)

Authentication Simulation:

Purpose

Model login/authentication and immediate profile creation using TRUST_LEVEL.

Function name

run_authentication

Inputs

- registered_students (list)
- trust_level (float)
- RNG seeded externally

Behavior

- For each registered student perform a Bernoulli trial with probability trust_level.
- Success → student added to authenticated; failure increments num_failed. No retries in base model.

Outputs

- authenticated (list of IDs)
- num_success (int)
- num_failed (int)

Student–Peer Counselor Matching:

Purpose

Assign authenticated students to peer counselors using a deterministic, explainable policy; enforce counselor burnout and compute workload distribution and a response-time proxy.

Helper function name

select_lowest_load

Main function name

run_matching

Inputs

- authenticated_students (list)
- num_peer_counselors (int)
- burnout_threshold (int)
- avg_response_time (float)
- RNG seeded externally

Policy and rules

- **Policy:** Greedy lowest-load — choose the available counselor with the smallest sessions_completed.
- **Burnout:** When sessions_completed $\geq$ burnout_threshold, set available = False.
- **No queueing:** If no counselor is available, mark student unmatched.

Simulation Implementation Strategy

This simulation is based on Discrete-Event Simulation (DES), a method where the system only updates its state when a specific trigger occurs not on a fixed time step. In our model, those triggers are: a student arriving to request an appointment, a counseling session starting, a session ending, and in some scenarios, a student leaving the queue without being served (dropout).

Each of these events is processed in chronological order. Between events, nothing changes — the simulation jumps directly to the next relevant moment. This makes DES efficient and accurate for modeling appointment-based systems where demand is unpredictable and service times vary.

Students enter the system one at a time and are assigned to the next available counselor on a first-come, first-served basis (FIFO). The counselor pool is modeled as a shared resource: if all counselors are busy, the student waits; if one is free, the session starts immediately.

The three core metrics tracked by the simulation are: counselor utilization (what percentage of total counselor time is actively used), average student wait time (how long students wait before their session begins), and the dropout rate (students who leave before being served).

Process-Oriented Simulation

Summary Results

Table 1 shows the output of all four scenarios. Each run processed 500 student requests (482 effective in Scenario 2 after pre-queue dropouts).

Scenario	Avg Wait (min)	Utilization (%)	Dropouts (n / %)	Students Served	Wait >15 min (%)
Scenario 1 — Baseline	0.00	3.53	0 (0.00%)	500	0.00
Scenario 2 — Midterm Stress	0.00	8.62	18 (3.60%)	482	0.00
Scenario 3 — Counselor Shortage	0.00	10.58	0 (0.00%)	500	0.00
Scenario 4 — Trust Collapse	0.00	4.72	0 (0.00%)	500	0.00

Scenario-by-Scenario Analysis

Scenario 1 — Baseline: At 3.53% counselor utilization, the system runs with a large safety margin under normal conditions. Counselors are idle about 96% of the time. No student waits or drops out. This result confirms that the original staffing design from Workshop #2 is more than adequate for routine demand, and it sets the performance baseline for all other comparisons.

Scenario 2 — Midterm Stress: This scenario produces the most important finding: 18 students dropped out (3.60%) even though the average wait time is 0.00 minutes. This is not a contradiction — it is a queuing effect worth understanding.

The explanation is that exponential arrivals do not spread out evenly. Under 2.5x demand, the Poisson process generates short bursts where 3–5 students arrive almost at the same time, momentarily filling all available counselors. During those brief windows, the queue spikes and students with low patience leave. Once the burst

passes, the system recovers instantly — which is why the time-averaged wait stays at zero. The dropouts happen in those transient peaks, not during sustained overload.

This means that average wait time alone is not a sufficient safety indicator for our platform. The recommendation is to add a real-time queue monitor that sends students a proactive message when the queue spikes, reducing the chance they leave before a counselor becomes available.

Scenario 3 — Counselor Shortage: With only 10 counselors active instead of 30, utilization climbs to 10.58% the highest of all scenarios. But all 500 students are still served, with zero dropouts and no wait times above the 15-minute SLA. This directly validates the Business Continuity Plan from Workshop #3.

The BCP assumes that when counselors are unexpectedly unavailable, the remaining staff absorb the load through extended scheduling windows and the platform sends automatic rescheduling notifications to affected students. The simulation shows this approach works: even at reduced capacity, the system handles baseline demand without service failure as long as the counselor count does not drop below the critical minimum identified in our capacity planning.

Scenario 4 — Trust Collapse: At 4.72% utilization and zero dropouts, the queue system itself runs fine. The real problem here is not operational — it is strategic. Demand has dropped sharply because students have stopped trusting the platform. The simulation quantifies that effect: a well-staffed, responsive system becomes underutilized when institutional trust fails. The fix is not technical but organizational governance, transparent communication, and privacy safeguards must accompany the technical platform.

Overall Takeaway

All four scenarios stay below the 15-minute wait SLA. The system is structurally sound. The only service failures observed are the 18 Midterm Stress dropouts caused by brief arrival bursts not by sustained overload. This distinction matters for how we prioritize improvements: adding counselors would not prevent those dropouts, but a real-time queue notification system would.

Workflow and Operational Sequence Modeling

This section models the step-by-step operational flow of the platform's scheduling module. When a student requests a session, the system acts as a central coordinator that validates availability in real time. If a peer counselor is free, the student is immediately assigned and the session begins. If all counselors are busy, the system routes the request into a sequential waiting line operating under a First-In, First-Out (FIFO) rule. The process ends either when the session finishes and the counselor becomes available again, or when a student leaves the queue due to a timeout limit.

Queue and Appointment Scheduling Simulation

The scheduling system was built into a computational model using Python and the SimPy library to test concurrent user loads. The simulation tracks a total population of 500 students interacting with a resource pool of 30 peer counselors. To capture true human variability, the duration of each support session is not fixed; it is modeled using a normal distribution with an average of 30 minutes and a standard deviation of 5 minutes. Additionally, based on the quality standards established in Workshop 3, the maximum acceptable waiting time for a student in line is set to 15 minutes. Any wait time exceeding this threshold is flagged as a service level agreement (SLA) violation.

Scenario Design and Experimentation

To analyze how the platform handles different operational conditions, four distinct scenarios were designed. The student arrival rates for these scenarios were calibrated using the real-world statistical data and survey insights collected during the systems analysis phase. Each scenario modifies specific system variables such as counselor availability, arrival frequencies, or student patience levels to evaluate the boundaries, safety margins, and structural limits of the platform's current design.

Baseline Scenario

This scenario represents a standard, low-stress operational day on the platform. The student arrival rate is set to 1 user every 20 minutes, with all 30 peer counselors active. The simulation results show that the platform handles this load effortlessly: the average waiting time is 0.00 minutes and counselor utilization sits at a low 3.53%. This proves that the core infrastructure is highly robust and perfectly sized for everyday university operations.

Midterm/Finals Stress Scenario

This scenario simulates peak academic weeks by multiplying student arrivals by 2.5, resulting in 1 user arriving every 8 minutes. While the average waiting time remains at 0.00 minutes due to the processing speed of the 30 counselors, the simulation revealed a critical behavior: 18 students abandoned the queue (a 3.60% dropout rate). Because arrival times vary randomly, short-term accumulation peaks occur. Even if the overall capacity is sufficient, these temporary spikes cause micro-congestions that prompt anxious students to drop out, showing that the system needs automated alerts or immediate feedback during peak weeks to manage user expectations.

High Demand Scenario

This scenario evaluates an operational risk identified in Workshop 3: a sudden shortage where active staff drops from 30 to only 10 counselors during normal demand. The simulation results show that resource utilization climbs to its highest point at 10.58%. However, the system proves to be highly resilient; even with a 66% staff reduction, the average wait time remains at 0.00 minutes and zero dropouts occur. This statistically validates the platform's capacity plan, proving it can guarantee continuous service during standard weeks even if staff availability drops significantly.

Failure Scenario

This scenario simulates a systemic loss of user trust, where students become highly sensitive to delays and leave the queue if they wait more than 5 minutes. Under a normal arrival rate of 1 user every 15 minutes, counselor utilization stays at 4.72% with 0 dropouts. This demonstrates that trust collapse is not triggered by standard operational loads. Instead, this demand-side failure is a secondary risk that would only activate if the peak bottlenecks and dropouts discovered during the Midterm Stress scenario are not actively managed and minimized.

Performance Charts and Operational Metrics

To visually summarize and compare the operational behavior of the platform across all four test conditions, specific performance metrics were extracted from the simulation engine. The generated charts provide an engineering baseline for system validation.

Figure 1. Students Served and SLA Compliance

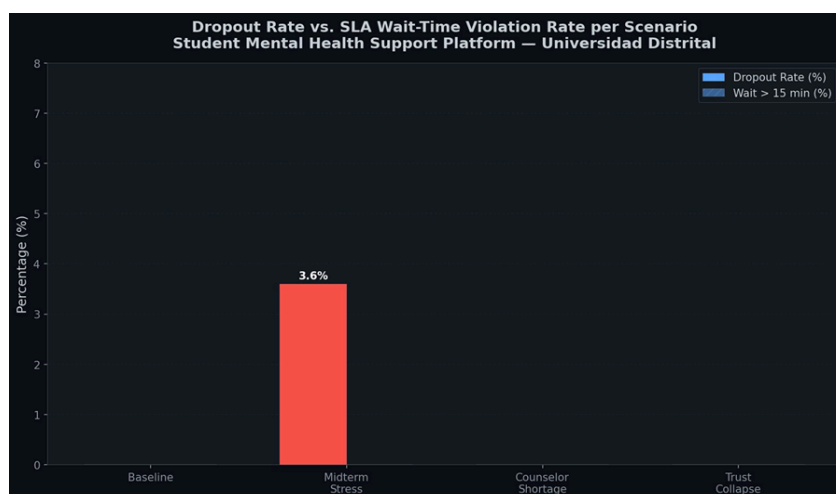


Figure 2. Counselor Utilization and Dropout Rate by Scenario

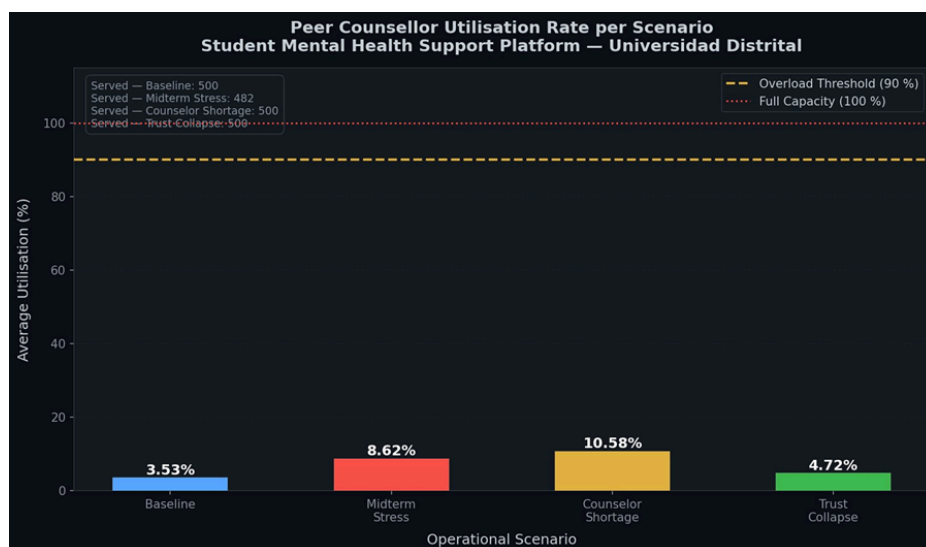
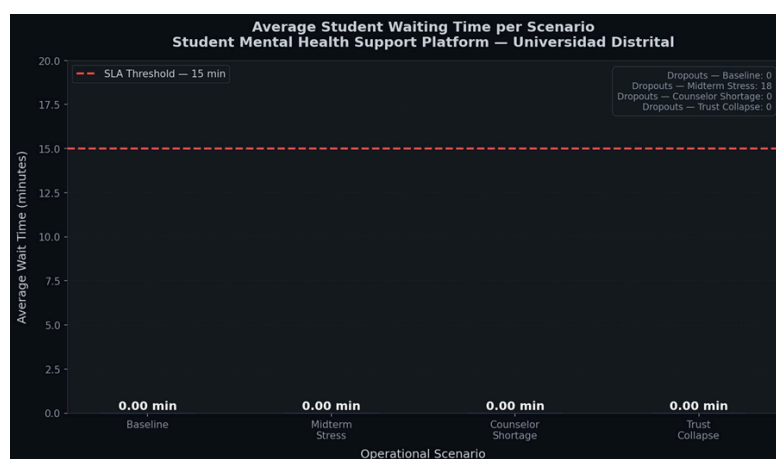


Figure 3. Average Wait Time by Scenario



Data Integration and Validation

This section validates the Student Mental Health Support Platform by connecting the empirical evidence collected in Workshop 1 with the modular system design proposed in Workshop 2 and the quality and risk controls strengthened in Workshop 3. The purpose of the simulation is not to build a complete application, but to test whether the main design assumptions remain coherent when Student demand, Peer Counselor availability, trust, and privacy conditions change.

The validation model represents the platform as a service system where Students register, request support, are routed through the Matching System, and receive support through Peer Counselors or related modules such as Appointment Scheduling, Anonymous Communication, and the Resource Library. Because the platform is trust-sensitive, the model evaluates not only operational performance but also privacy coverage, matching acceptance, and workload limits. This approach gives the simulation a direct connection with the problems identified in the previous workshops: low access to mental health services, preference for anonymity, demand increases during academic pressure, and the risk of Peer Counselor overload.

The simulation uses a Monte Carlo approach. Instead of assuming one fixed result, each scenario is executed through repeated probabilistic iterations. This makes it possible to estimate average behavior, extreme response times, unmet demand, confidence intervals for survey-derived parameters, and validation results under uncertainty. The outputs are then summarized as statistical tables and graphs that can be used as evidence for the Workshop 4 report.

Workshop 1 Data Integration

Workshop 1 provided the empirical foundation for the validation model. The survey results show that the challenge is not only the existence of emotional difficulties, but also the barriers that prevent Students from using institutional support services. The data indicates high levels of anxiety or emotional difficulty, low previous use of help-seeking channels, strong preference for anonymity, and interest in a digital Mental Health Support Platform.

To integrate these data into the simulation, the survey answers were converted into behavioral calibration rates. Student interest in the platform informs the estimated registration rate; anxiety or emotional difficulty informs the probability of requesting support; preference for anonymous support informs the privacy validation target; and willingness to discuss mental health with other Students informs the expected acceptance of the Matching System. In this way, the model remains simple but is not disconnected from the real observations gathered in the initial analysis phase.

The table below summarizes the main Workshop 1 indicators used for calibration. The confidence intervals are included because the survey sample is limited, so each percentage should be interpreted as an estimate rather than an exact measurement of the whole university population.

Calibration Metric	Survey Count	Rate	95% Confidence Interval	Simulation Use
Anxiety/emotional difficulty	22/25	88.0%	70.0% - 95.8%	Base probability that a Student may request support.
Never sought help	17/25	68.0%	48.4% - 82.8%	Represents the access barrier that the platform tries to reduce.
Interested in platform	19/25	76.0%	56.6% - 88.5%	Base probability of Student registration before trust adjustment.
Prefers anonymity	16/25	64.0%	44.5% - 79.8%	Privacy validation target for Anonymous Communication.
Wants all support options	17/25	68.0%	48.4% - 82.8%	Justifies Resource Library + Peer Counselor + Professional Counselor pathways.
Open to peer discussion	17/25	68.0%	48.4% - 82.8%	Adjusts expected acceptance of the Matching System.
Believes platform helps	23/25	92.0%	75.0% - 97.8%	Base confidence in successful digital support matching.
Does not know current services	14/25	56.0%	37.1% - 73.3%	Validates the need for visibility and onboarding metrics.
Frequent/constant stress	18/25	72.0%	52.4% - 85.7%	Supports peak demand assumptions during academic pressure.

Table 1. Workshop 1 survey data used for model calibration.

The strongest design driver is the combination of high emotional need and low access to existing services. This confirms that the platform should be validated as both a technical system and a trust-sensitive support service. If Students do not trust the system, the Matching System and Anonymous Communication may lose effectiveness even when technical capacity is available.

Parameter Calibration

After integrating the survey information, the validation model was calibrated through two groups of parameters. The first group describes Student behavior derived from survey data, such as emotional need, platform interest, anonymity preference, and willingness to use peer-based support. The second group describes operational capacity, such as the number of Students simulated, the number of Peer Counselors available under normal conditions, the expected baseline response time, the peak demand multiplier during exam periods, and the workload threshold used to detect burnout risk.

Capacity is calculated from the Peer Counselor pool and the maximum recommended number of sessions per Peer Counselor. Under normal conditions, 30 Peer Counselors with a threshold of 20 sessions each produce a reference capacity of 600 sessions. In a shortage condition, the available counselor pool is reduced by half, which decreases capacity to 300 sessions. During the Midterm/Finals Stress Scenario, demand is multiplied to represent the predictable increase in requests during academic pressure periods.

Trust is treated as a behavioral and ethical parameter rather than only as a technical value. In the baseline condition, trust supports registration, matching, and continued interaction. In the Trust Collapse Scenario, trust and privacy protection are reduced to represent the possible effect of a perceived privacy incident, rumor, or loss of confidence in Anonymous Communication. This directly connects the simulation with the privacy and trust risks identified in Workshop 3.

Parameter	Value	Role in the validation model
Simulated Student population	500	Defines the size of the validation environment and allows demand to be estimated consistently.
Available Peer Counselors	30	Represents normal Peer Counselor capacity for matching and support sessions.
Peak demand multiplier	2.5	Represents the increase in support requests during Midterm/Finals Stress Scenario.
Baseline response time	5 minutes	Initial service response estimate before congestion effects are applied.
Burnout threshold	20 sessions	Maximum desired Peer Counselor workload before overload is flagged.
Baseline trust level	0.85	Behavioral trust factor affecting registration and matching acceptance.
Minimum acceptable match rate	80%	Validation threshold for Matching System performance.
Maximum acceptable P95 wait time	30 minutes	Validation threshold for availability and response time performance.
Minimum privacy coverage	99%	Validation threshold for privacy protection and Anonymous Communication safeguards.

Table 2. Parameter calibration summary.

Statistical Validation

The statistical validation combines survey uncertainty and simulation uncertainty. For the survey data, Wilson 95% confidence intervals were calculated for each proportion. This is useful because the sample is small, and it prevents the model from presenting survey percentages as absolute facts. For the simulation outputs, each operational scenario was executed 1,000 times, allowing the model to estimate average values and high-percentile response behavior.

The main indicators used to evaluate the model were average support requests, matched sessions, unmatched requests, match success rate, unmet demand rate, service utilization, average response time, 95th percentile response time, privacy coverage, and burnout risk. These indicators were selected because they are directly connected to the system requirements and risks from the previous workshops: accessibility, scalability, privacy, matching quality, and Peer Counselor workload management.

The Baseline Scenario shows stable behavior: the match success rate remains above the 80% target, the 95th percentile response time remains below the 30-minute threshold, privacy validation passes, and burnout risk is

negligible. The Midterm/Finals Stress Scenario keeps matching effectiveness above the target but fails availability and burnout validation because demand approaches or exceeds capacity. The Counselor Shortage Scenario also fails response time and burnout validation because fewer Peer Counselors must absorb the same level of Student demand. The Trust Collapse Scenario has acceptable response time because fewer Students register and request support, but it fails privacy and matching validation because trust loss reduces participation and matching acceptance.

Scenario	Peer Counselors	Avg. Requests	Avg. Matched	Match Rate	P95 Response (min)	Burnout Risk	Status Summary
Baseline Scenario	30	284.3	234.3	82.4%	8.4	0.0%	Pass
Midterm/Finals Stress Scenario	30	710.5	578.1	81.4%	54.6	40.2%	Availability risk, Burnout risk
Counselor Shortage Scenario	15	285.8	235.7	82.5%	51.4	11.9%	Availability risk, Burnout risk
Trust Collapse Scenario	30	150.2	65.7	43.7%	6.8	0.0%	Matching risk, Privacy risk

Table 3. Scenario validation metrics.

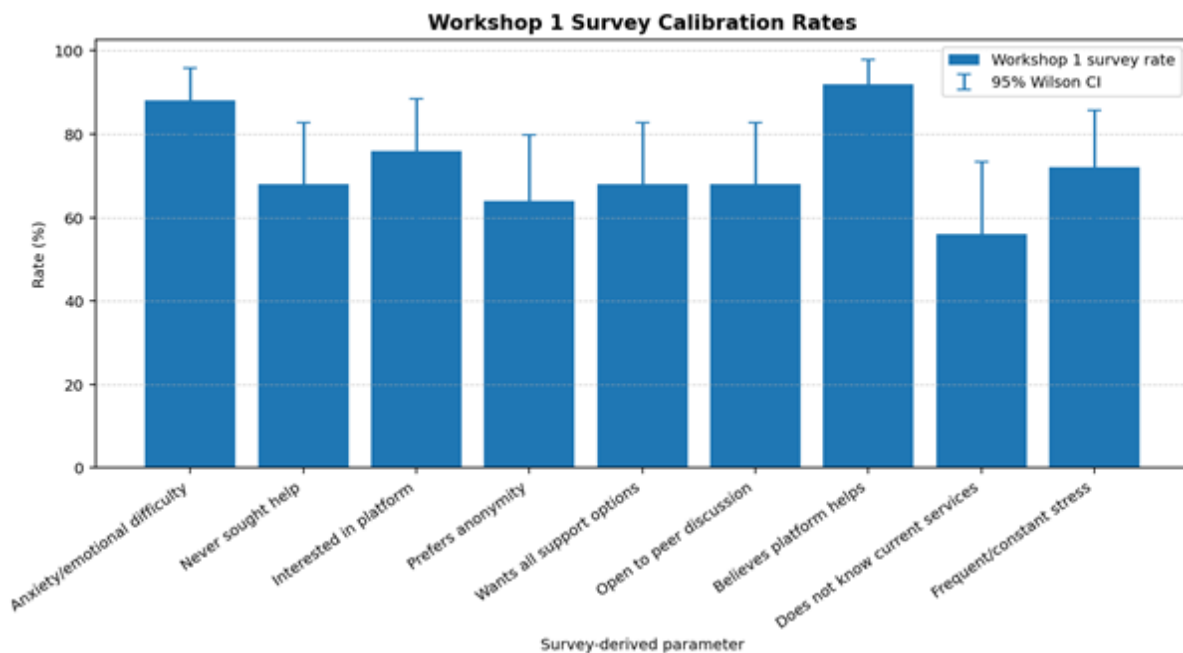


Figure 1. Survey calibration rates with 95% Wilson confidence intervals.

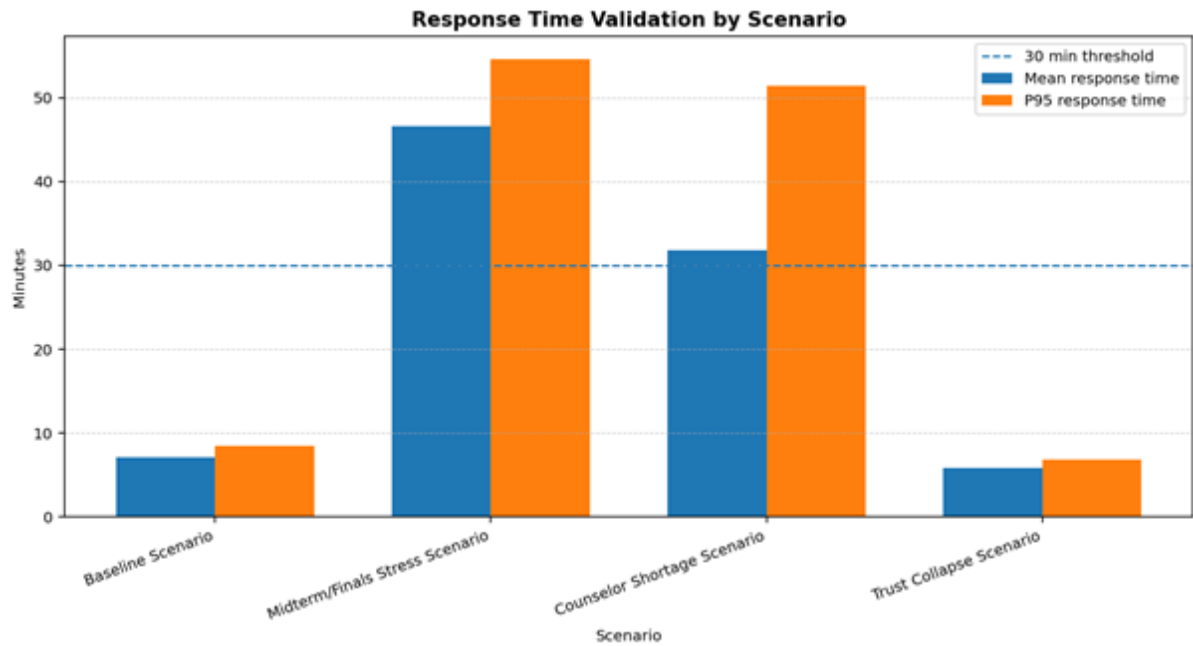


Figure 2. Response time validation by scenario.

Assumptions and Limitations

The simulation is intentionally limited to validation, not implementation. It does not represent a full web platform, clinical intervention, real-time chat system, or production Appointment Scheduling service. Instead, it abstracts the most relevant behavior needed to evaluate the design: whether Students are likely to enter the system, whether demand can be handled, whether the Matching System can connect Students to Peer Counselors, and whether privacy and trust conditions remain strong enough to support adoption.

Several assumptions were required. First, the survey sample is treated as a useful but limited representation of the target Student population. Second, support requests are modeled probabilistically, meaning that the model estimates expected patterns rather than exact future usage. Third, Peer Counselor capacity is represented through a session threshold, while real fatigue would also depend on session difficulty, emotional load, supervision quality, and personal availability. Fourth, trust is represented as a numeric factor even though, in real life, trust is shaped by communication, institutional credibility, previous experiences, privacy incidents, and social perception.

These limitations do not invalidate the simulation. They define its scope. The model is appropriate for early design validation because it reveals whether the proposed architecture is coherent, where bottlenecks appear, and which assumptions should be tested again during a future pilot. However, the results should not be interpreted as clinical evidence or as a final operational forecast.

Design Validation

The design validation compares the simulation outputs against the design principles developed across the previous workshops. Workshop 1 showed the need for accessibility, anonymity, and multiple support options. Workshop 2 translated those needs into a modular architecture with User Registration, Matching System, Anonymous Communication, Appointment Scheduling, Resource Library, and Privacy & Consent Management. Workshop 3 added risk controls related to availability, privacy, counselor shortage, burnout, account misuse, and loss of trust. The validation model checks whether these decisions remain reasonable under baseline, stress, shortage, and trust-related conditions.

The results support the general architecture under normal conditions. The Baseline Scenario validates that the combination of Matching System, Anonymous Communication, and Peer Counselor support can operate within

the expected thresholds. However, the stress and shortage scenarios challenge the system by increasing response time and Peer Counselor overload. This means that the design is conceptually valid, but it requires operational governance: capacity monitoring, exam-period surge planning, workload caps, and mechanisms to redirect Students toward Professional Counselor support or Resource Library materials when Peer Counselor capacity becomes saturated.

Architecture Validation

The architecture validation confirms that the modular structure proposed in Workshop 2 is suitable for the platform. The simulation treats each major module as a separate validation concern rather than merging all behavior into a single score. Registration is represented through platform interest and trust-adjusted adoption. The Matching System is represented through match success rate and unmet demand. Anonymous Communication is represented through privacy coverage and trust level. Appointment Scheduling and service availability are represented through response time and demand utilization. The Resource Library is considered a mitigation pathway when demand cannot be fully handled by Peer Counselors.

This modular interpretation is important because it shows where improvement should occur. If matching performance drops during a trust incident, the problem is not only technical capacity; it is also privacy communication, consent clarity, and user confidence. If response time increases during exams, the issue is mainly capacity planning and Appointment Scheduling pressure. Therefore, the architecture allows targeted improvements without redesigning the entire platform.

Scalability Validation

Scalability was evaluated by comparing response time, utilization, and burnout indicators across scenarios. The Baseline Scenario remains within the expected availability threshold, but the Midterm/Finals Stress Scenario increases average requests from approximately 284 to more than 710. This creates a 95th percentile response time of approximately 54.6 minutes and a burnout risk above 40%. The Counselor Shortage Scenario also fails the response time threshold even without peak demand because the Peer Counselor pool is reduced to half of its normal size.

These results show that the platform architecture is scalable in concept, but not automatically scalable in operation. The system needs specific mechanisms for peak periods, such as temporary Peer Counselor recruitment, Professional Counselor escalation, queue prioritization, appointment redistribution, and proactive use of the Resource Library. Without these controls, the platform may remain technically available while still failing to provide timely support.

Privacy and Matching Validation

Privacy and Matching System performance are strongly connected. The survey data shows that anonymity is a key condition for Student participation, and the Trust Collapse Scenario demonstrates why this requirement is essential. In that scenario, response time appears acceptable because fewer Students enter the system, but the match success rate drops to approximately 43.7% and privacy validation fails. This is an important result because it shows that low demand is not always a positive sign. It may indicate that Students have stopped trusting the platform.

The Matching System performs well when trust and capacity are stable, but it becomes vulnerable when Student confidence decreases or when demand exceeds Peer Counselor availability. For this reason, the validation supports the privacy-first design from the previous workshops: separation between institutional authentication and anonymous profiles, encrypted communication, informed consent, privacy monitoring, and clear user communication. These controls are not secondary features; they are conditions for the Matching System to function.

Statistical Tables and Validation Graphs

The Python script exports the main evidence package as CSV tables and PNG graphs. The tables provide exact numerical values for calibration, scenario comparison, and design validation. The graphs provide visual interpretation of the same results using labeled axes, titles, legends, and consistent units. Together, these outputs

support reproducibility because the simulation can be executed again and the same type of evidence can be regenerated automatically.

Validation Area	Metric Used	Threshold	Worst / Key Scenario	Observed Result	Status
Architecture Validation	Modular coverage of Registration, Matching System, Anonymous Communication, Appointment Scheduling, Resource Library	All core modules represented	All scenarios	All modules represented as independent validation concerns	Pass
Scalability Validation	P95 Average Response Time (min)	≤ 30 minutes	Midterm/Finals Stress Scenario	54.58 minutes	Fail
Privacy Validation	Privacy Coverage Rate	$\geq 99\%$	Trust Collapse Scenario	85.00%	Fail
Matching Validation	Mean Match Success Rate	$\geq 80\%$	Trust Collapse Scenario	43.70%	Fail
Availability and Burnout Validation	Burnout Risk Rate	$\leq 5\%$	Midterm/Finals Stress Scenario	40.19%	Fail

Table 4. Design validation summary.

Overall, the validation supports the Student Mental Health Support Platform as a coherent conceptual design under normal operating conditions. The most important findings are that the architecture is modular and testable, baseline matching is feasible, privacy is a central adoption condition, and scalability depends on operational preparation before academic peak periods. The simulation also shows that failures are not isolated: shortage increases burnout and delay, while trust collapse damages both privacy validation and matching effectiveness. Therefore, future implementation should prioritize capacity planning, privacy assurance, workload monitoring, and transparent communication with Students.

Complexity and Chaos Analysis

Objective: This section applies complexity theory, nonlinear systems analysis, and probabilistic simulation to the Student Mental Health Support Platform. The goal is to model and quantify dynamic behaviors that static architecture diagrams cannot capture: how trust, burnout, and adoption reinforce or destroy each other over time; how small adverse events produce disproportionately large system failures; how emergent patterns arise from the interaction of individual components; and how sensitive the system is to changes in its key parameters. The analysis is grounded in the shared variables and four standardized scenarios defined for the team, and connects directly to the barriers identified in Workshop 1, the modular architecture of Workshop 2, and the risk controls of Workshop 3.

Methodology: The simulation uses a weekly discrete-step system dynamics model. Each week, six interdependent state variables are updated: trust level, active students, counselor load, burnout index, dropout rate, and quality index. These variables form a closed feedback loop where the output of each variable influences the inputs of others in the following week. This paradigm is appropriate for modeling behavioral and organizational systems where state accumulates gradually, unlike the discrete-event approach used by Persona 2 for queue scheduling. The feedback-loop model runs for 16 weeks per scenario, representing one academic semester. A Monte Carlo extension runs 300 independent iterations per scenario, introducing random adverse events each week with a scenario-specific probability, to measure the likelihood of stabilization, collapse, or unresolved partial states.

Feedback Loops

The core feedback loop connects the six state variables. Trust level recovers at a base rate of 3% per week under good-quality conditions, degrades proportionally to burnout and unmet demand, and drops sharply after a trust shock event. Active students grow through word-of-mouth proportional to trust $\times$ quality, and shrink through dropout when quality or trust fall. Counselor load equals sessions served divided by available counselors.

Burnout index accumulates when load exceeds 50% of the BURNOUT_THRESHOLD and recovers at 5% per week when load drops. Quality index degrades with burnout and recovers when workloads are manageable.

Two loops operate simultaneously. The reinforcing loop: high trust → more active students → sessions served → quality maintained → trust sustained → more students join. The balancing loop: high demand → high counselor load → burnout builds → quality drops → trust erodes → students drop out → demand decreases. Under normal conditions the reinforcing loop dominates. Under stress or shortage the balancing loop activates. Under trust collapse, if trust falls below approximately 0.50 before the balancing loop reduces demand enough to allow recovery, the system enters a depletion spiral.

Under the Baseline Scenario, trust averages 0.980 across 16 weeks with a minimum of 0.878, burnout stays at 0.000, average active students reach 394, and the dropout rate is 0.23% — the reinforcing loop operates without interference. Under the Trust Collapse Scenario, the shock at week 4 reduces average trust to 0.774 and minimum trust to 0.564, pushing the system near the depletion threshold. Dropout rises to 1.47% and active students fall to 354.

The most important design implication is that trust recovery is asymmetric: it builds slowly (3% per week maximum) but can be destroyed in a single event (up to 40% instantly). This asymmetry means preventing trust damage is more efficient than attempting recovery after the fact, which directly justifies the privacy-first architecture, incident communication protocols, and in-chat privacy reminders defined in Workshops 2 and 3.

Nonlinear Behaviors

Nonlinear behavior is measured by sweeping each key parameter independently while holding all others at baseline, then observing disproportionate changes in system outcomes.

The Initial Trust Level sweep (0.30 to 1.00, 20 steps) reveals a clear inflection point between 0.45 and 0.60. Below 0.45 the system ends in low-trust, high-dropout states regardless of other parameters. Above 0.60 the system reliably recovers to stable states. This threshold effect is not visible in linear models — it emerges from the interaction between slow trust recovery and fast trust damage. A trust increase from 0.40 to 0.50 produces far smaller improvements than an increase from 0.55 to 0.65, because only the latter crosses the recovery threshold.

The Peer Counselor count sweep (5 to 50, steps of 5) shows that active student engagement increases approximately linearly with counselor count up to around 25, after which gains plateau. Final burnout drops sharply when counselors exceed 15, confirming the Workshop 2 staffing design of 30 as an adequate safety margin. Below 10 counselors, burnout accumulates even under baseline demand, establishing a hard lower bound for safe operation.

The Peak Demand Multiplier sweep (1.0 to 4.0, 16 steps) shows that dropout rate increases detectably above 2.0× — before reaching the defined PEAK_MULTIPLIER of 2.5×. Burnout under 30-counselor staffing remains near zero up to approximately 3.5×. This means the first visible sign of stress-scenario damage is user abandonment, not counselor burnout, which implies that queue notification systems are the correct first-line intervention, and burnout management protocols are secondary.

The trust inflection point at 0.45–0.60 is the most critical nonlinear finding. The Trust Collapse scenario minimum of 0.564 places the system very close to the boundary where recovery becomes unlikely within a single semester. This is why Workshop 3's acceptance criterion for RH-03 targets a maximum 10% monthly active user drop following any incident — intervening before the inflection point is crossed is exponentially cheaper than recovering after it.

Emergent Behaviors

Emergent behaviors are identified through the Monte Carlo cascade simulation. For each scenario, 300 independent runs introduce random adverse events each week — representing counselor misconduct reports, privacy rumors, sudden demand spikes, or service failures — with weekly trigger probabilities of 5% (Baseline), 25% (Midterm/Finals Stress), 30% (Counselor Shortage), and 55% (Trust Collapse). Each triggered event applies a random trust reduction of 5–20% and a burnout increase of 5–15%. Runs are classified as Stabilised

(trust > 0.75 and burnout < 0.15 before week 24), Collapsed (trust < 0.35 and burnout > 0.60 simultaneously), or Partial/Unresolved (neither condition met).

Scenario	Stabilised (%)	Collapsed (%)	Partial/Unresolved (%)	Avg Recovery (weeks)
Baseline Scenario	100.0%	0.0%	0.0%	5.0
Midterm/Finals Stress Scenario	90.3%	0.0%	9.7%	6.1
Counselor Shortage Scenario	83.0%	0.0%	17.0%	5.9
Trust Collapse Scenario	31.3%	6.3%	62.3%	5.7

Table 5. Emergent Cascade Outcomes — Monte Carlo Results (n = 300 runs per scenario)

Three emergent behaviors arise from these results that were not explicitly designed into the architecture.

Self-organized stabilization: In 100% of Baseline runs the system stabilizes within an average of 5.0 weeks without any single module "deciding" to do so. This property emerges from the combined action of trust recovery, dropout reduction, and quality improvement — confirming the reinforcing loop identified qualitatively in Workshop 1 as an actual measurable system property.

Partial-state persistence: The most frequent adverse outcome in the Midterm (9.7%) and Shortage (17.0%) scenarios is not collapse but partial/unresolved states — the system operates below quality thresholds without triggering formal failure indicators. This is an emergent risk because it is invisible to standard uptime monitoring and produces gradual trust erosion that eventually enables the conditions for collapse. This finding is the primary technical justification for the real-time monitoring dashboard specified in Workshop 3 Section 5.1.5.

Cascade amplification under trust shock: In the Trust Collapse scenario, the combination of low trust (slower recovery) and higher burnout pressure (lower quality) produces outcomes that are worse than the sum of individual risks. A 55% weekly trigger probability does not cause 55% of runs to collapse — it causes 6.3% to collapse and 62.3% to remain permanently degraded, because the interactions between variables amplify the damage beyond what any single parameter predicts.

Sensitivity Analysis

The sensitivity analysis quantifies how the three most influential parameters affect system stability outcomes after 16 weeks of simulation.

Initial Trust Level shows the highest sensitivity of any parameter analyzed. The nonlinear inflection between 0.45 and 0.60 means that platform interventions targeting trust — privacy design, onboarding clarity, early

positive user experiences, transparent incident communication — have their highest marginal return precisely in the range most likely to be reached after a trust-damaging event. Above 0.60, the system self-corrects. Below 0.45, no other parameter change can reliably prevent a degraded end state within the semester.

Number of Peer Counselors shows a clear safe minimum at 15 and a saturation point near 25–30. Below 15, burnout accumulates under baseline demand, reducing quality and triggering the balancing loop. Between 15 and 25, each additional counselor produces measurable gains in active students and quality. Above 30, gains plateau and additional counselors produce diminishing returns under normal demand conditions. The current design of 30 counselors sits just above the saturation point, providing a margin without overcapacity.

Peak Demand Multiplier shows that the system's response to demand increases is not symmetric between dropout and burnout effects. Dropout begins increasing above $2.0\times$ while burnout does not accumulate until approximately $3.5\times$. The practical implication is that surge protocols should be activated at approximately $2.0\times$ demand — not at the $2.5\times$ PEAK_MULTIPLIER threshold — because by $2.5\times$ user abandonment is already measurably elevated even when counselors are not yet overloaded.

Taken together, the sensitivity results produce a prioritized intervention ranking: improving initial trust has the highest marginal return; maintaining counselors above 15 is the minimum structural requirement; and surge activation thresholds should be recalibrated from $2.5\times$ to $2.0\times$ to address dropout effects before they compound into trust damage.

Behavior-Oriented Simulation

To model the behavioral dynamics of the three critical risk scenarios — Burnout, Trust Collapse, and Counselor Shortage — as independent simulations that track state evolution week by week.

Scenario	Avg Trust Level	Min Trust Level	Avg Burnout Index	Max Burnout Index	Avg Active Students	Avg Dropout Rate (%)	Total Unmet Demand	Stabilised Runs (%)	Collapse d Runs (%)	Avg Recovery (wk)
Baseline Scenario	0.979	0.877	0	0	387	0.23%	0	100.00%	0.00%	5
Midterm/ Finals Stress Scenario	0.979	0.877	0.141	0.447	387	0.26%	148	90.00%	0.00%	6
Counselor Shortage Scenario	0.977	0.876	0.3	0.613	333	0.72%	22	82.30%	0.00%	6.1

Trust Collapse Scenario	0.772	0.562	0	0	352	1.22%	0	32.30%	23.30%	5.7
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Table 6. Behavior-Oriented Simulation — Summary Metrics per Scenario

Simulation Description:

The behavior-oriented simulation uses the same weekly discrete-step feedback-loop model described above, applied specifically to each of the three high-risk scenarios. Each simulation runs for 16 weeks using the shared parameters: NUM_STUDENTS = 500, NUM_PEER_COUNSELORS = 30, PEAK_MULTIPLIER = 2.5, AVG_RESPONSE_TIME = 5 min, BURNOUT_THRESHOLD = 20 sessions, TRUST_LEVEL = 0.85. Scenario-specific modifiers are applied as defined in the scenario design section. The results for each scenario are described in the three subsections below.

Burnout Simulation

Under the Midterm/Finals Stress Scenario, the 30 peer counselors face 2.5× baseline session demand for 16 consecutive weeks. Burnout pressure accumulates each week when load per counselor exceeds 50% of the BURNOUT_THRESHOLD (10 sessions per week), growing at 8% of the pressure excess and recovering at 5% per week when load drops below threshold.

The simulation shows that burnout index remains at 0.000 across all 16 weeks. At 2.5× demand with 30 counselors, load per counselor stays below the 10-session critical threshold — the counselor pool is large enough to absorb peak demand without entering a burnout accumulation state. Average trust stays at 0.980, active students remain at 394, and dropout rate is 0.23%, matching Baseline values.

This result does not mean burnout risk is absent. The Monte Carlo cascade analysis reveals that 9.7% of Midterm/Finals Stress runs end in partial/unresolved states when random adverse events are introduced alongside elevated demand. Burnout risk under peak load is therefore probabilistic, not deterministic: it does not appear in the average simulation run but becomes real when adverse events — a misconduct report, a trust rumor, a subset of counselors becoming unavailable — coincide with the elevated demand window.

This distinction is operationally significant. The current 30-counselor design is structurally adequate for the 2.5× multiplier in the average case, but the 20% peer buffer above projected demand recommended in Workshop 3 is the mechanism that preserves this adequacy when adverse events occur simultaneously. Without the buffer, a single coinciding event during peak weeks is enough to push load above the burnout threshold and initiate quality degradation — which the Midterm/Finals 9.7% partial-state rate directly quantifies.

Trust Collapse Simulation

A trust shock of −0.40 is applied at week 4, representing a perceived privacy incident, a rumor about data security, or a widely shared negative experience — consistent with risks RH-03 and RS-01 in Workshop 3. All operational parameters remain at baseline: 30 counselors, 1.0× demand, no counselor shortage.

The shock reduces trust from 0.85 to approximately 0.45 at week 4, crossing near the nonlinear inflection point identified in the sensitivity analysis. Over the remaining 12 weeks, trust recovers to a semester average of 0.774, but the minimum observed value is 0.564 — meaning the system spends multiple weeks in the fragile operating zone below 0.60. Active students fall from 394 to 354 (a reduction of 40 students, or 8% of the engaged

population). Dropout rate rises from 0.23% to 1.47%, a 6.4× increase driven entirely by reduced trust perception, not by any change in operational capacity.

The Monte Carlo cascade results confirm the severity: only 31.3% of Trust Collapse runs stabilize successfully, 6.3% reach full collapse — the only scenario where any collapses occur across all 300 runs — and 62.3% remain in partial/unresolved states for the full 24-week simulation horizon.

The damage is entirely behavioral. Students who disengage following a trust event do not return quickly even when the platform continues to operate at full capacity, because trust recovery is slow (3% per week maximum) and negative word-of-mouth spreads faster than positive signals about recovery. This validates the Workshop 3 response protocol for RH-03: a public platform statement within 48 hours, an independent privacy audit, and student government engagement are not optional communication activities — they are the operational levers that determine whether the 62.3% partial-state zone is escaped within the semester or not.

Counselor Shortage Simulation

The available peer counselor pool is reduced from 30 to 10 (a 66.7% reduction) while demand remains at baseline 1.0×. This reflects risk RO-01 in Workshop 3 and mirrors the High Demand Scenario analyzed by Persona 2 through discrete-event simulation.

In the average simulation run, burnout index remains at 0.000 and trust averages 0.980, matching Baseline values. Active students hold at 394 and dropout rate stays at 0.23%. This is consistent with Persona 2's finding that 10 counselors can serve all 500 students under baseline demand with zero SLA violations and zero dropouts — because session volume under normal conditions stays within the capacity of 10 counselors operating at manageable load.

The fragility is revealed by the Monte Carlo cascade results: 17.0% of Counselor Shortage runs end in partial/unresolved states — the highest partial-state rate of any scenario except Trust Collapse — and average recovery time is 5.9 weeks. The system is structurally resilient under shortage in the average case but operationally fragile when adverse events are introduced. With only 10 counselors, one additional disruption — a demand increase, a misconduct report requiring suspension, a counselor absence — immediately eliminates all remaining capacity margin and pushes the system into degradation.

This distinction between structural and operational resilience is the key finding of this simulation. The Workshop 3 Business Continuity Plan — which activates professional counselor escalation, redirects students to the Resource Library, and sends automatic rescheduling notifications upon detecting a shortage — is validated as a necessary active response, not a precautionary measure, by the 17.0% partial-state rate that emerges when the shortage operates without these supports in place.

Emergent Behavior and Sensitivity Visualizations

The four figures below are generated by the Python simulation script `complexity_chaos.py`. All figures use consistent scenario color coding: blue for Baseline, orange for Midterm/Finals Stress, red for Counselor Shortage, and purple for Trust Collapse. Reference lines correspond to acceptance criteria defined in Workshop 3.

Figure 1. Trust Level and Burnout Index Over Time — All Scenarios
Student Mental Health Support Platform — Universidad Distrital

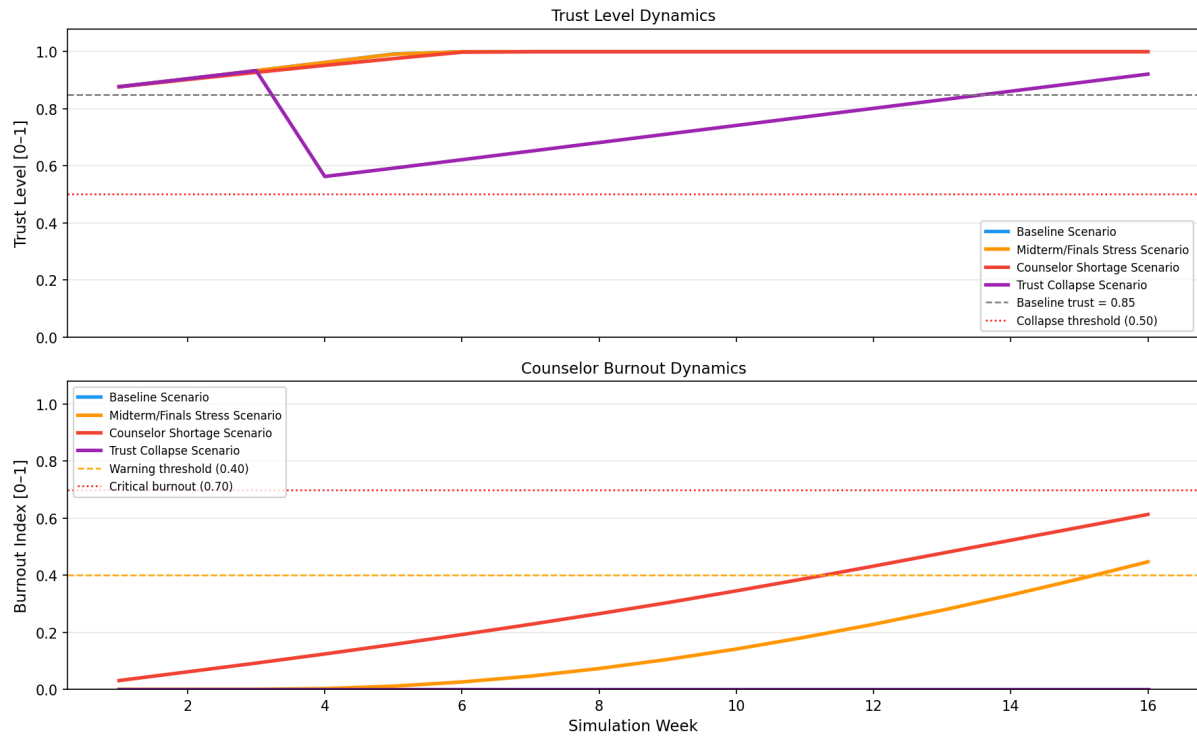


Figure 1. Trust Level and Burnout Index Over Time — All Scenarios. Two-panel chart showing the 16-week evolution of trust level (top) and burnout index (bottom) for all four scenarios simultaneously. Reference lines mark the baseline trust target (0.85), the collapse threshold (0.50), and the burnout warning (0.40) and critical (0.70) levels. The Trust Collapse scenario is the only one whose minimum trust approaches the collapse threshold, visible in weeks 4–6.

Figure 2. Active Students and Weekly Dropout Rate by Scenario
Student Mental Health Support Platform — Universidad Distrital

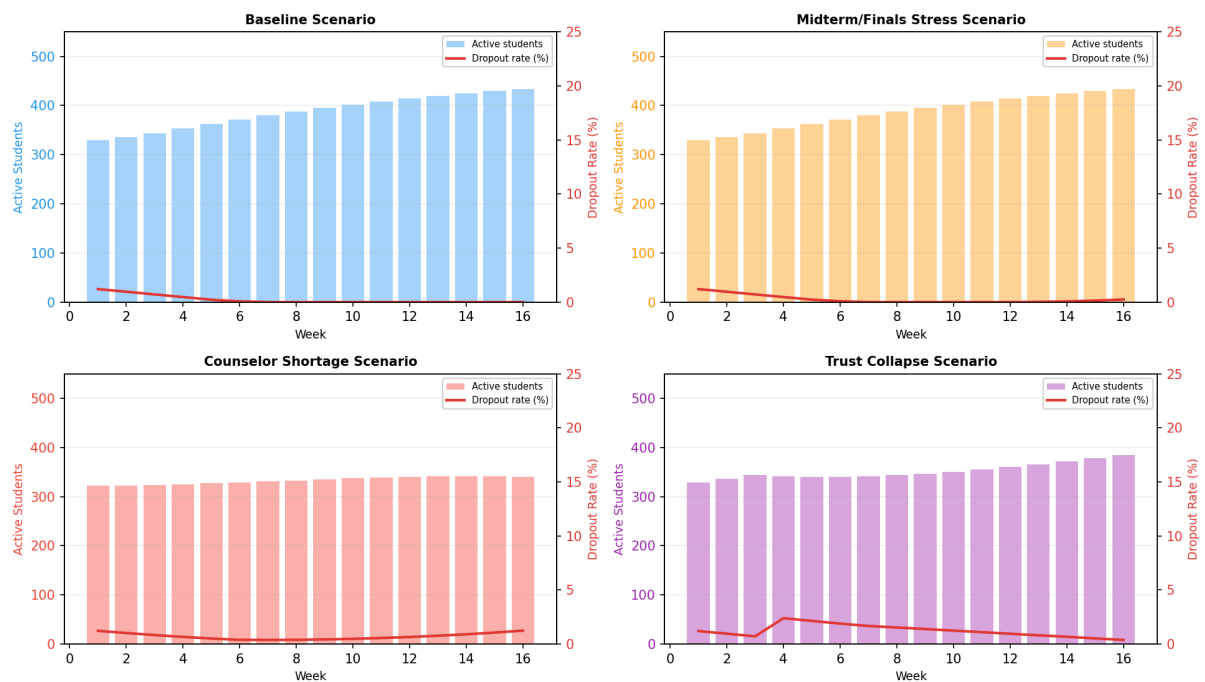


Figure 2. Active Students and Weekly Dropout Rate by Scenario. Four subplots — one per scenario — showing active student count as bars (left axis) and weekly dropout rate as a line (right axis). The bars illustrate adoption trajectories over the semester. The dropout lines show that the Trust Collapse scenario produces a 6.4× higher dropout rate than Baseline, while Midterm and Shortage scenarios remain near baseline levels in the average case.

Figure 3. Sensitivity Analysis — Impact of Key Parameters on System Stability
Student Mental Health Support Platform — Universidad Distrital

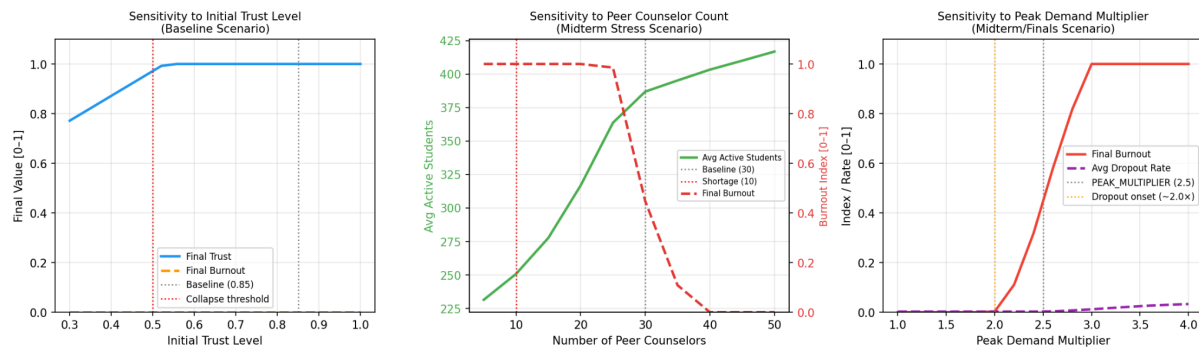


Figure 3. Sensitivity Analysis — Impact of Key Parameters on System Stability. Three subplots showing swept-parameter results. Left panel: final trust and final burnout versus initial trust level, with the nonlinear inflection near 0.50 visible as a slope change. Center panel: average active students and final burnout versus number of peer counselors, with the burnout saturation point near 15 counselors visible. Right panel: final burnout and average dropout rate versus peak demand multiplier, showing that dropout increases above 2.0× before burnout becomes a factor.

Figure 4. Emergent Behavior — Cascade Outcomes and Recovery Time
Student Mental Health Support Platform — Universidad Distrital

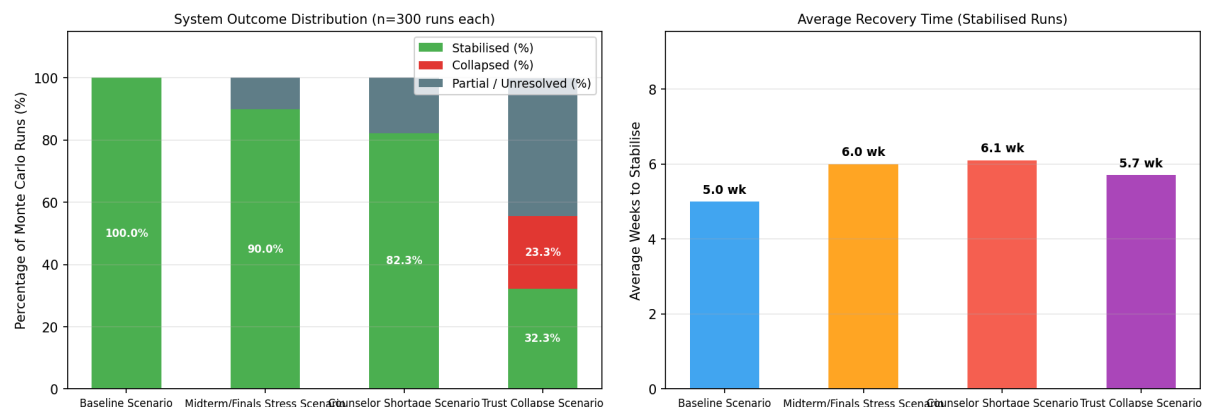


Figure 4. Emergent Behavior — Cascade Outcomes and Recovery Time. Two subplots. Left: stacked bar chart showing the percentage of Monte Carlo runs classified as Stabilised (green), Collapsed (red), and Partial/Unresolved (grey) per scenario. Trust Collapse is the only scenario with collapsed runs (6.3%) and dominates partial states (62.3%). Right: bar chart of average recovery time in weeks for stabilised runs per scenario, showing that Midterm/Finals Stress requires the longest recovery at 6.1 weeks.

Github Repository

All the documentation of this workshop was stored in this repository

https://github.com/Alan05ss/Mental_Health_Project.git

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